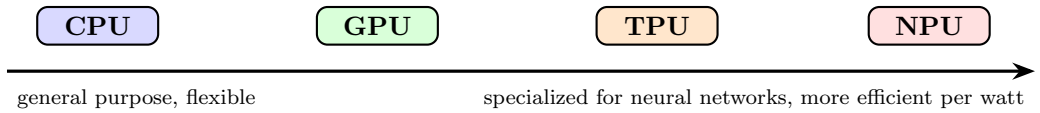
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CPU, GPU, TPU and NPU

Setting: all four are processors, but they trade **flexibility** for **efficiency** differently. The more a chip is specialized for one kind of math (the matrix multiplications behind neural networks), the more operations per second and per watt it delivers, and the fewer kinds of programs it runs well.



1. CPU (Central Processing Unit): the general-purpose brain of every computer. **Few** (about 4–128) **powerful cores**, each with large caches, branch prediction and out-of-order execution, optimized for **low latency on sequential, branchy code**: operating system, databases, web servers, data loading, orchestration. It can run neural networks, but slowly, since it does relatively few multiply-adds in parallel. *Examples:* Intel Xeon / Core, AMD EPYC / Ryzen, ARM (AWS Graviton, Apple M-series cores).

2. GPU (Graphics Processing Unit): designed for graphics, where millions of pixels get the same computation. **Thousands of simple cores** run the same instruction on different data (SIMT), plus **Tensor Cores** specialized in small matrix multiplications, and very high memory bandwidth (HBM, terabytes per second). Optimized for **throughput**, not latency. This matches deep learning almost perfectly, so the GPU is the **default hardware for training and serving** models, with the most mature software ecosystem (CUDA, cuDNN, PyTorch). *Examples:* NVIDIA H100 / H200 / B200, AMD Instinct MI300.

3. TPU (Tensor Processing Unit): an ASIC (application-specific chip) designed by **Google** only for neural networks. Its core is a **systolic array**: a large grid of multiply-accumulate units through which data flows rhythmically, so each value read from memory is reused many times without going back to memory. For a matrix product $C = A \cdot B$, each cell computes

$$c_{ij} \leftarrow c_{ij} + a_{ik} \cdot b_{kj} \quad \text{as } a \text{ flows right and } b \text{ flows down through the grid,}$$

which minimizes memory traffic, the main energy cost. Uses low precision formats such as **bfloat16** and INT8, and scales to **pods** of thousands of chips linked by a fast dedicated interconnect for training very large models. Available only through **Google Cloud**; programmed via XLA (JAX, TensorFlow, PyTorch/XLA). Less flexible than a GPU for unusual operations.

4. NPU (Neural Processing Unit): a small AI accelerator **inside** the system-on-chip of a phone, laptop, car or camera. Built for **on-device inference** at very **low power** (a few watts), using quantized

arithmetic (INT8, INT4): face unlock, photo enhancement, speech recognition, live translation, background blur, small local LLMs. Benefits: privacy (data stays on the device), no network latency, offline use and battery savings. Not used for training large models. *Examples:* Apple Neural Engine, Qualcomm Hexagon NPU, Intel and AMD NPUs in "AI PCs", Google Tensor in Pixel phones.

	CPU	**GPU**	**TPU**	**NPU**
Cores	Few, powerful	Thousands, simple + Tensor Cores	Large systolic arrays	Small MAC arrays
Optimized for	Latency, sequential logic	Parallel throughput	Matrix math at scale	Inference per watt
Flexibility	Any program	High (CUDA)	Neural networks only	Neural network inference only
Precision	FP64 / FP32	FP32 to FP8 / FP4	bfloat16, INT8	INT8, INT4
Power	Tens to hundreds of W	Hundreds of W to $\sim$1 kW	Hundreds of W	A few W
Where	Every device and server	Data centers, workstations, gaming PCs	Google Cloud only	Phones, laptops, edge devices
ML role	Data prep, classic ML, orchestration	Training + serving (default)	Large-scale training + serving	On-device inference

How they work together: in a training cluster, **CPUs** load and preprocess data and coordinate, while **GPUs or TPUs** do the matrix math; the trained model is then quantized and deployed to an **NPU** to run locally on a phone or laptop. A rule of thumb: CPU for logic, GPU for flexible parallel math, TPU for massive matrix math on Google Cloud, NPU for efficient AI at the edge.